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1. (Special note: Below there are five questions for you to answer. Naturally there are many mathematical formulas and equations in the problem statements. However, due to some technical issues on Coursera, from time to time for some learners, some formulas and equations cannot be displayed correctly (while some others can). As there is no way for the instructing team to solve this issue, we post all the problem statements in a PDF file [here](#) [🔗](#). If unfortunately some formulas and equations are not readable for you, please use the PDF file to prepare your answers and then come back to Coursera to choose/fill in your answers.)
- 1 / 1 point

According to the instructor, the CSR scheduling problem discussed in this week exhibits which of the following characteristics? Check all correct answers.

- ☒ Allocation.
- ☒ Selection.
- ☐ Balancing.
- ☐ Sequencing.
- ☐ All of the above.

2. Which of the following statements best describes the main decision to make in the CSR scheduling problem discussed in this week?
- 1 / 1 point
- ☐ For each CSR, determine her/his number of days off in the coming month.
- ☐ For each day, determine the number of CSRs that should go to work.
- ☐ For each day, determine the number of CSRs in each shift.
- ☒ For each CSR for each day, determine her/his shift.
- ☐ None of the above.
- ☒ Correct

3. What is the difference between a hard constraint and a soft constraint?
- 1 / 1 point
- ☐ A hard constraint can be formulated with mathematical expressions while a soft one cannot.
- ☐ A hard constraint does not have a decision variable in it while a soft one does.
- ☒ A hard constraint must be satisfied while a soft one is not.
- ☐ All of the above.
- ☐ None of the above.
- ☒ Correct

4. Recall that in the CSR scheduling problem discussed in this week, we want to minimize the maximum number of night shifts among all CSRs is considered. Suppose that now instead we want to minimize the maximum number of night-shift CSRs among all work days (let w_3 be such a number), which of the following are the correct set of constraints to calculate the number?
- 1 / 1 point
- ☒ $w_3 \geq 0, w_3 \geq \sum_{i \in I} \sum_{k \in K^N} x_{ijk}$ for all $j \in J$.
- ☐ $w_3 \geq 0, w_3 \geq \sum_{j \in J} \sum_{i \in I} \sum_{k \in K^N} x_{ijk}$.
- ☐ $w_3 \geq 0, w_3 \geq \sum_{i \in I} \sum_{j \in J} x_{ijk}$ for all $k \in K^N$.
- ☐ $w_3 \geq 0, w_3 \geq \sum_{k \in K^N} x_{ijk}$ for all $j \in J, i \in I$.
- ☐ None of the above.
- ☒ Correct

5. The technique for solving the CSR scheduling problem discussed in this week is the most appropriate for solving which of the following problem?
- 1 / 1 point
- ☐ Determining the daily production quantities of all products for a manufacturer.
- ☐ Determining the allocation of investment budget among stocks.
- ☐ Determining the salary for each employee.
- ☒ Determining how to assign multiple jobs to multiple workers in multiple days.
- ☐ Determining the restaurant for the dating in the coming Christmas eve.
- ☒ Correct